Lab 8: Later subagents re-run searches the first ones already finished

Topic 1: Agentic Architecture & Orchestration · Task 1.3: Configure subagent invocation, context passing, and spawning

Runs on Simulator 3: Multi-Agent Research System · Self-practice

Source: SkillCertPro practice exam Part 1, Q35 (unofficial third-party question)

The question

After a web-exploration phase in a Claude Code research workflow, later subagents repeatedly rerun searches that earlier subagents already completed and fail to preserve why certain sources were rejected. The coordinator receives brief outputs from the earlier subagents, but next-phase work still starts as if no exploration happened. What should the coordinator change before spawning each next-phase subagent?

A. Spawn the next subagents normally and rely on their isolated context windows to access the parent history.

B. Save the prior findings for the final synthesis prompt and keep next subagent delegations independent.

C. Condense prior exploration into a source-linked summary and include it in each next subagent's delegation prompt.

D. Attach complete prior subagent transcripts to every next delegation so no detail is omitted.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 8
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 8

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py --case UC-35
```

What you should see

Synthesis and report prompts carry the source links, far smaller than the raw subagent traffic.

Synthesis and report prompts contain the source links found earlier and are far smaller than the raw subagent traffic.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

C. Add a source-linked summary of prior phases to each delegation prompt.

A normal subagent starts with a fresh, isolated context, so it does not know what earlier subagents found, rejected or left open. After each phase, write a summary with key findings, source links, rejection reasons and unresolved questions, and put it in the next delegation prompt. That matches the documented chained-subagent pattern.

Why the others fail

A. An isolated context window is precisely what prevents access to the parent's history.

B. Intermediate subagents need that state too, or they duplicate work and lose provenance-sensitive decisions.

D. Spends large amounts of context on low-value exploration output.

Check yourself

- synthesis packet contains >=3 source links/doc names from phase 1
- report packet contains the synthesis output (>=1500 chars)
- no packet exceeds 60000 characters (no full transcripts)

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.